

Kinematic Analysis of Existing 5-Axis FFF Architectures and Proposal of a CoreXY-Based Distributed-Rotation System

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Abstract— Conventional fused filament fabrication (FFF) systems are typically limited to three translational degrees of freedom, constraining deposition to planar layers and thereby introducing well-known limitations in support generation, surface finish, and direction-dependent mechanical properties. Multi-axis extrusion systems have been developed to address these limitations by adding rotational degrees of freedom, but existing 5-axis architectures exhibit important tradeoffs in kinematic complexity, structural stiffness, calibration sensitivity, and reachable deposition orientation for larger build plate sizes. This paper first examines existing 5-axis FFF system architectures from a robotics perspective, with particular emphasis on the previously proposed 3-axis gantry + 2-axis rotary-stage configuration. Using a spatial kinematic framework, the analysis considers the relative nozzle–part pose, the role of coordinate-frame assignment, and the implications of this head–table architecture for workspace, singularity behavior, and geometric error propagation. This review highlights the strengths of rotary-stage systems, including mechanical simplicity and translational stiffness, while also identifying limitations associated with stage-centered orientation, collision constraints, and amplified sensitivity to rotary misalignment. Motivated by these observations, the paper then introduces a new 5-axis FFF architecture based on a Cartesian CoreXY platform with two distributed rotary degrees of freedom: one rotational axis associated with the build-frame plane and a second rotational axis located at the toolhead. In contrast to conventional rotary-stage systems that place both additional degrees of freedom on the workpiece, the proposed design distributes orientation capability across the machine structure and deposition subsystem. This arrangement is intended to preserve the accessibility and rigidity of a Cartesian mechanism while expanding local nozzle orientation control. A preliminary kinematic formulation of the proposed system is presented to establish the basis for future analysis of workspace, singularities, calibration, and process-feasible toolpath generation. More broadly, the work positions the proposed architecture as a candidate middle ground between conventional Cartesian printers, head–table 5-axis systems, and fully articulated robotic additive manufacturing platforms.

Keywords— First word or key phrase, second word or key phrase, third word or key phrase. (Place between three and six key words or phrases separated by a comma, which represent the theme of your work)

I. INTRODUCTION

Fused deposition modeling (FDM), or more generally fused filament fabrication (FFF), is widely used because of its relatively low cost, open hardware ecosystem, and straightforward material handling. However, conventional desktop FDM systems are typically restricted to three translational degrees of freedom, depositing material in planar layers stacked along a fixed build direction. This layer-wise constraint creates several well-known limitations, including visible stair-stepping on curved surfaces, direction-dependent mechanical properties, and the frequent need for support structures when fabricating overhangs or geometrically complex parts [1, 2, 3].

Multi-axis extrusion-based additive manufacturing has emerged as a response to these limitations. By introducing additional rotational degrees of freedom, the deposition direction can be reoriented relative to the part geometry, enabling conformal layer deposition, improved access to overhanging regions, and greater control over filament alignment. Prior reviews of multi-degree-of-freedom additive manufacturing have shown that these systems are motivated not only by geometric freedom, but also by process-level goals such as support reduction, improved surface quality, and more favorable anisotropic mechanical behavior [1, 3]. In this broader context, so-called “5-axis” FDM systems are better understood as robotic deposition systems in which nozzle pose relative to the part is no longer constrained to a fixed vertical build direction.

Among the possible 5-axis machine layouts, the *3-axis gantry + 2-axis rotary-stage* configuration is especially attractive for research and prototyping. In this architecture, the printhead retains conventional Cartesian motion in the

x -, y -, and z -directions, while the workpiece is reoriented by two additional rotary axes, typically implemented as a tilt-rotation stage or gimballed build platform. Compared with serial robotic-arm systems, this arrangement offers a simpler kinematic structure, lower cost, and higher structural stiffness in the translational axes, while still allowing the relative nozzle-part orientation to vary during fabrication [2, 4]. Open-access systems such as *Open5x* further demonstrate that this architecture can be realized as an upgrade path from a conventional desktop printer, making it particularly relevant for experimental robotics and low-cost machine design research [4, 5].

From a robotics perspective, the significance of this architecture lies in the fact that it behaves as a *head-table* manipulator rather than as a purely Cartesian machine. The Cartesian gantry defines the nozzle position in a machine reference frame, while the rotary stage changes the pose of the part through a sequence of rigid-body rotations. As a result, the deposition process depends on the relative transform between the nozzle frame and the part frame, rather than on translational coordinates alone. This introduces many of the same core questions found in robot kinematics: frame assignment, forward and inverse kinematic mapping, workspace feasibility, singular or near-singular orientations, and the propagation of geometric calibration errors into end-effector pose error [6, 7]. In particular, because the rotary axes are offset from the deposition point, even small angular misalignment can produce significant Cartesian positioning errors at the nozzle, making calibration and compensation central to machine performance [7].

The mechanics of the 3-axis gantry + 2-axis rotary-stage layout also distinguish it from other multi-axis extrusion systems. By preserving a largely conventional printhead and filament path, the architecture avoids some of the compliance and packaging issues associated with aggressively tilted or elongated non-planar printheads. At the same time, relocating two degrees of freedom to the build platform introduces new challenges related to stage stiffness, backlash, inertial loading, and collision between the tilted part and the surrounding machine envelope [3, 4]. These tradeoffs make the architecture an important middle ground between fully articulated robotic-arm deposition and traditional planar FDM: it is sufficiently rich to require formal kinematic treatment, yet sufficiently structured to permit tractable modeling and calibration.

In parallel with mechanical and kinematic developments, recent work in multi-axis extrusion has shown that toolpath generation cannot be separated from machine geometry. Five-axis process planning must account for reachable nozzle orientations, collision-free part poses, and process constraints such as bead support, local surface normal alignment, and continuous extrusion stability [6, 8]. Open-source slicer workflows such as those developed for *Open5x* and multidirectional slicing systems further suggest that practical 5-axis printing often relies on constrained orientation strategies rather than unrestricted arbitrary nozzle motion [4, 5]. Consequently, a rigorous study of this machine class should begin with its kinematic structure, since reachable orientations and geometric error propagation directly shape what toolpaths are physically realizable.

Accordingly, this work (paper/publication?) focuses on the kinematic analysis of a 3-axis gantry + 2-axis rotary-stage FDM system. The goal is to model the machine as a five-degree-of-freedom deposition robot, define the relevant coordinate frames, derive the forward kinematic relationship between the gantry and rotary-stage subsystems, and establish a basis for evaluating workspace, orientation capability, and calibration sensitivity. By treating the printer as a robotic mechanism rather than only as a manufacturing device, the analysis aims to clarify how this architecture balances accessibility, stiffness, and orientation freedom, and why it remains one of the most compelling platforms for low-cost multi-axis extrusion research.

II. KINEMATIC MODEL OF A 3-AXIS GANTRY + 2-AXIS ROTARY-STAGE PRINTER

A 3-axis gantry + 2-axis rotary-stage printer can be modeled as a five-degree-of-freedom (5-DOF) *head-table* robotic system. In this architecture, the printhead retains conventional Cartesian motion in the x -, y -, and z -directions, while the workpiece is reoriented by a two-axis rotary stage. This arrangement is attractive because it preserves the mechanical simplicity and stiffness of a conventional gantry while adding sufficient orientation freedom for conformal and support-reduced deposition (i'm using deposition a lot synonym?) [2, 4]. Opensource implementations such as *Open5x* explicitly follow this conversion approach by augmenting a standard Cartesian printer with a controllable two-axis rotating stage [4, 5]. At the same time, recent calibration work has shown that low-cost 5-axis printers should be treated as fully kinematic machines, since rotary-axis misalignment and offset errors can significantly deform the intended fabricated component (workpiece?) [7]. For clarity, the *Open5x* printer design is presented in figure 1.

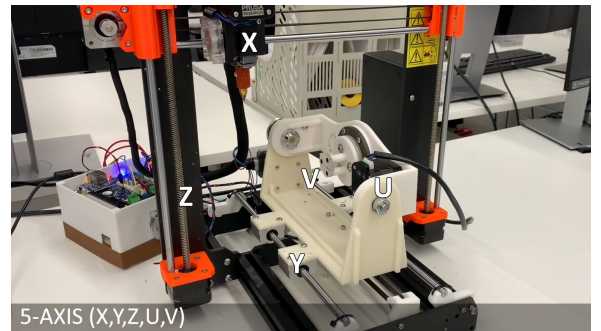


Fig. 1: *Open5x* 5-axis printer architecture shown at approximately 15 s in the project video. The system is labeled with three Cartesian gantry axes (X, Y, Z) and two rotary-stage axes (U, V), illustrating the head-table configuration analyzed in this work. Image captured from the *Open5x* demonstration video [9], with system context and terminology consistent with prior descriptions of the *Open5x* platform in [4, 10].

a. Frame Definitions and Notation

Let the following right-handed coordinate frames be defined:

- $\{W\}$: world or machine base frame,
- $\{N\}$: nozzle frame,

- $\{G\}$: gantry carriage reference frame,
- $\{S_0\}$: base frame of the rotary stage,
- $\{S_1\}$: intermediate frame after the first rotary joint,
- $\{P\}$: part or build-platform frame.

The principal quantity of interest is the nozzle pose relative to the part,

$${}^P\mathbf{T}_N = ({}^W\mathbf{T}_P)^{-1} {}^W\mathbf{T}_N, \quad (1)$$

where ${}^A\mathbf{T}_B \in SE(3)$ denotes the homogeneous transformation from frame $\{B\}$ to frame $\{A\}$. Equation (1) makes explicit that deposition depends on the *relative* pose between the nozzle and the part, rather than on gantry position alone. This is a defining feature of multi-axis extrusion systems and is one reason that 5-axis printing must be analyzed as a robotic mechanism rather than only as a conventional machine tool [6, 4].

b. Cartesian Gantry Subsystem

The gantry subsystem is modeled as three orthogonal prismatic joints. Let x , y , and z denote the commanded translations along the machine axes. Then the nozzle pose in the world frame may be written as

$${}^W\mathbf{T}_N = \mathbf{T}_x(x) \mathbf{T}_y(y) \mathbf{T}_z(z) {}^G\mathbf{T}_N, \quad (2)$$

where ${}^G\mathbf{T}_N$ is the fixed transform from the gantry carriage reference point to the nozzle tip. The translational transforms are

$$\mathbf{T}_x(x) = \begin{bmatrix} 1 & 0 & 0 & x \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{T}_y(y) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (3)$$

$$\mathbf{T}_z(z) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & z \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

This subsystem is often conceptually described as a *PPP* chain. However, unlike a classical serial manipulator, the gantry is more naturally represented directly in the machine frame through orthogonal translations than through a strict Denavit–Hartenberg (DH) construction.

c. Rotary-Stage Subsystem

The rotary stage is modeled as a two-revolute-joint (*RR*) serial chain. Let the stage coordinates be denoted by A and B , representing the first and second rotary joints, respectively. The part pose in the world frame is

$${}^W\mathbf{T}_P = {}^W\mathbf{T}_{S_0} {}^{S_0}\mathbf{T}_{S_1}(A) {}^{S_1}\mathbf{T}_P(B), \quad (4)$$

with ${}^W\mathbf{T}_{S_0}$ the fixed installation transform from the machine base to the stage base.

A DH-style representation of the two revolute joints can be written using the standard homogeneous transform

$$\mathbf{A}_i(\theta_i, d_i, a_i, \alpha_i) = \mathbf{R}_z(\theta_i) \mathbf{T}_z(d_i) \mathbf{T}_x(a_i) \mathbf{R}_x(\alpha_i), \quad (5)$$

where θ_i is the joint angle, d_i is the offset along the joint axis, a_i is the common-normal distance, and α_i is the twist angle between adjacent joint axes. For the two-axis stage,

$${}^W\mathbf{T}_P = {}^W\mathbf{T}_{S_0} \mathbf{A}_4(A, d_4, a_4, \alpha_4) \mathbf{A}_5(B, d_5, a_5, \alpha_5) {}^{S_2}\mathbf{T}_P. \quad (6)$$

A corresponding rough DH parameter table is

i	θ_i	d_i	a_i	α_i
4	A	d_4	a_4	α_4
5	B	d_5	a_5	α_5

(7)

For many practical rotary-stage designs, the two axes are intended to be approximately orthogonal, so $\alpha_i \approx \pm\pi/2$. If the axes nearly intersect, then a_i may be small; otherwise, nonzero a_i and d_i capture the true mechanical offset between the two rotary axes. These parameters are especially important in low-cost systems, where assembly tolerances and printed mechanical parts can produce substantial deviations from the nominal geometry [7].

d. Combined Spatial Kinematic Model

Combining the gantry and the stage yields the full head–table model. Substituting (2) and (4) into (1) gives

$${}^P\mathbf{T}_N = ({}^W\mathbf{T}_{S_0} {}^{S_0}\mathbf{T}_{S_1}(A) {}^{S_1}\mathbf{T}_P(B))^{-1} (\mathbf{T}_x(x) \mathbf{T}_y(y) \mathbf{T}_z(z) {}^G\mathbf{T}_N). \quad (8)$$

Equation (8) provides a compact representation of the printer as a 5-DOF deposition robot:

$$\underbrace{PPP}_{\text{gantry}} + \underbrace{RR}_{\text{rotary stage}}. \quad (9)$$

This representation is useful because it separates the translational positioning task from the orientation task while still capturing their coupling through the relative transform. In inverse-kinematic terms, a desired nozzle pose relative to the part is realized by selecting stage angles (A, B) to achieve an admissible local orientation, followed by solving for the corresponding gantry position (x, y, z) .

e. Geometric Sensitivity and Error Amplification

A key feature of this mechanics model is that rotary-axis errors are amplified with distance from the stage center. If a deposition point is located at radial distance r from a stage axis, then for a small angular error $\delta\theta$ (in radians), the induced Cartesian error magnitude is approximately

$$\delta s \approx r \delta\theta. \quad (10)$$

Thus, even modest angular misalignment in the rotary stage can create significant nozzle-placement error at points far from the pivot. This behavior has been emphasized in recent calibration work on low-cost 5-axis printers, where kinematic identification and compensation were required to maintain acceptable deposition accuracy [7]. It also explains why 5-axis printers require full geometric calibration rather than only conventional bed leveling while also having a greatly reduced build volume compared to traditional Core-XY or Rectilinear Motion Systems [11].

f. Singularity Challenges of the Head–Table Model

Although the 3-axis gantry + 2-axis rotary-stage printer is mechanically simpler than a 6-DOF serial robot arm, it still exhibits singularity-related behavior. In this context, singularity should be understood broadly as a loss of favorable local mapping between commanded joint motion and realized nozzle–part pose.

Let the generalized coordinate vector be

$$\mathbf{q} = [x \ y \ z \ A \ B]^T. \quad (11)$$

The differential kinematics may be written as

$$\dot{\mathbf{x}} = \mathbf{J}(\mathbf{q}) \dot{\mathbf{q}}, \quad (12)$$

where $\dot{\mathbf{x}}$ is a differential representation of relative nozzle–part motion and $\mathbf{J}(\mathbf{q})$ is the system Jacobian. Singular or near-singular configurations arise when $\mathbf{J}$ loses rank or becomes poorly conditioned.

For the head–table architecture, the main singularity challenges are as follows.

Rotary-axis alignment singularities. If the two stage axes approach geometric alignment in the transformed frame, then one rotational degree of freedom becomes locally redundant. This is analogous to gimbal-lock behavior in orientation parameterizations: two nominally distinct stage motions produce nearly the same change in part orientation. Near such configurations, orientation control becomes ill-conditioned, and small changes in desired orientation can require disproportionately large joint motions.

Workspace-boundary singular behavior. As the part tilts toward the mechanical limits of the stage, the system may remain kinematically solvable but lose practical controllability. The tilted part can approach the nozzle body, gantry frame, or enclosure, and the available collision-free orientation space collapses. In this regime, the printer exhibits a form of boundary singularity in which mathematically admissible poses are no longer manufacturable. Open5x explicitly notes that bed size and collision angle must be managed to reduce interference between the tilted build plate and the nozzle [4].

Process singularities. There are also deposition-specific singularities not captured by rigid-body kinematics alone. A theoretically reachable nozzle orientation may be unusable because the filament feed path, nozzle geometry, gravity-assisted bead support, or cooling conditions become unfavorable. As a result, the admissible workspace is smaller than the purely geometric workspace. This is consistent with five-axis process-planning literature, which treats kinematic feasibility, collision avoidance, and local deposition conditions as coupled constraints rather than independent stages [6].

g. Obtuse-Angle Process Singularity of a Fixed Vertical Toolhead

A further limitation arises when the nozzle remains fixed in a vertically oriented configuration while only the part is

rotated. In this case, the machine may possess nominal rotational degrees of freedom, yet still encounter configurations in which the local deposition direction cannot follow the surface normal beyond a critical angle. A fixed vertically oriented toolhead therefore introduces an *obtuse-angle process singularity*, in which the part may continue to rotate, but the nozzle can no longer achieve a process-feasible orientation relative to the local surface. In particular, for strongly curved or spherical geometries (such as a hemisphere or something <- change DO NOT KEEP), there exist regions in which the local surface tangent becomes effectively obtuse relative to the fixed nozzle axis. Once this occurs, the reachable set of admissible nozzle–surface orientations collapses because the nozzle itself cannot tilt to maintain a favorable deposition angle.

As a result, the deposition direction collapses toward nearly horizontal layer placement with respect to gravity and the machine frame, regardless of further rotation of the part. At the same time, the physical nozzle body, heater block, or surrounding toolhead geometry may intersect the part before a useful surface-normal alignment is reached. This is especially problematic for spherical or highly convex surfaces, where the outer curvature causes the fixed vertical toolhead to approach the part from an increasingly unfavorable direction. In such cases, the limiting factor is not the existence of a mathematical joint solution, but the loss of a collision-free and process-feasible deposition pose.

This singularity also highlights an important manufacturing limitation of fixed-vertical multi-axis systems: even when part rotation enables some conformal improvement, it does not fully eliminate conventional layer-line artifacts on curved surfaces. This is illustrated in figure 2 with a hemisphere decomposed into manufacturable layerlines using the OrcaSlicer 3D Printing software.

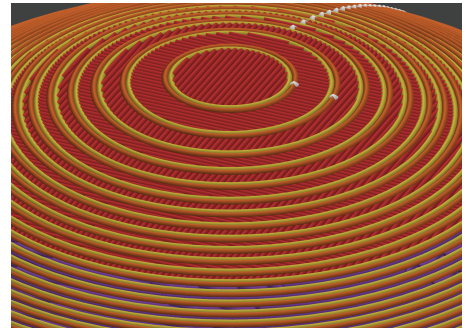


Fig. 2: Author-generated OrcaSlicer visualization of a hemispherical part, showing layer-line behavior associated with the obtuse-angle process singularity of a fixed vertical toolhead. The image illustrates how deposited layers remain orthogonal or near-orthogonal to the curved surface over significant regions, leading to persistent staircase artifacts.

Because the nozzle orientation remains constrained, deposition is still dominated by layers whose effective bead cross-sections remain orthogonal, or nearly orthogonal, to the local curved geometry over significant regions. Consequently, visible layer lines may be reduced in limited areas, but they are not fundamentally eliminated on continuously curved features. In other words, the inability of the toolhead to align with changing surface normals prevents full conformal deposition, so staircase effects

persist wherever the deposited layers intersect the curved surface at non-tangential angles.

h. Implications for Kinematic Analysis

These singularity mechanisms suggest that a complete analysis of the 3-axis gantry + 2-axis rotary-stage printer should not stop at deriving forward kinematics. In addition to the transform model in (8), the system should be evaluated in terms of:

1. reachable orientation workspace,
2. Jacobian conditioning over the build volume,
3. sensitivity of nozzle pose to rotary-axis misalignment,
4. collision-constrained stage angles, and
5. process-feasible deposition orientations.

Accordingly, the kinematic model developed here provides the foundation for later analysis of workspace, calibration sensitivity, and orientation planning. The central challenge of this machine class is not merely adding two rotational axes, but managing the coupled spatial kinematics that arise when a Cartesian gantry is combined with a reorientable part frame.

III. EXPLICIT SINGULARITY ANALYSIS FOR OBTUSE-ANGLE PROCESS SINGULARITY OF A FIXED VERTICAL TOOLHEAD AND ROTARY-AXIS ALIGNMENT SINGULARITIES

This section is going to be a more mathematically rigorous analysis of Rotary-axis alignment singularities and Obtuse-Angle Process Singularity of a Fixed Vertical Toolhead. This will lead into my proposed model and how it addresses these 2 issues specifically

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IV. PROPOSED COREXY-BASED 5-AXIS ARCHITECTURE

Motivated by the limitations of fixed-vertical and rotary-stage-only 5-axis extrusion systems, a new architecture is proposed in which rotational capability is distributed directly within the tool-side kinematic chain. The mechanism consists of a Cartesian CoreXY gantry providing planar translation in x and y , a vertical translational axis in z , a first rotary joint located at the connection between the tool assembly and the X -gantry, and a second rotary joint located at the toolhead itself. A finite rigid link of length L separates the two rotary axes. This arrangement allows the deposition direction to be modified through tool-side articulation rather than relying exclusively on reorientation of the part.

From a robotics perspective, the system may be interpreted as a hybrid kinematic chain composed of a

translational Cartesian base and a two-revolute serial wrist-like subsystem,

$$\underbrace{PPP}_{\text{CoreXY} + z} + \underbrace{RR}_{\text{tool-side rotations}}. \quad (13)$$

Unlike a gantry + rotary-stage architecture, in which the part frame is reoriented while the nozzle remains fixed, the proposed design changes nozzle orientation directly. As a result, the relative nozzle-part pose can be controlled through both translational motion of the gantry and rotational motion of the tool chain, potentially improving accessibility to curved features and reducing the orientation limitations associated with a fixed vertical toolhead.

a. Coordinate Frames and Notation

Let the following right-handed coordinate frames be defined:

- $\{W\}$: machine base or world frame,
- $\{G\}$: gantry carriage frame,
- $\{J_1\}$: frame attached to the first rotary joint at the connection to the X -gantry,
- $\{J_2\}$: frame attached to the second rotary joint at the toolhead,
- $\{N\}$: nozzle frame,
- $\{P\}$: part frame.

The generalized coordinate vector is defined as

$$\mathbf{q} = [x \ y \ z \ \theta_1 \ \theta_2]^T, \quad (14)$$

where x , y , and z are the Cartesian gantry coordinates, θ_1 is the rotation at the joint connecting the tool assembly to the X -gantry, and θ_2 is the rotation at the toolhead. The rigid distance between the two rotary joints is denoted by L .

As in the previous analysis, the principal quantity of interest is the nozzle pose relative to the part,

$${}^P\mathbf{T}_N = ({}^W\mathbf{T}_P)^{-1} {}^W\mathbf{T}_N, \quad (15)$$

where ${}^A\mathbf{T}_B \in SE(3)$ denotes the homogeneous transformation from frame $\{B\}$ to frame $\{A\}$.

b. Cartesian Base Motion

The translational subsystem is modeled as a Cartesian motion stage. Although the planar motion is physically realized through a CoreXY belt arrangement, the end-effector kinematics are still equivalent to independent Cartesian translation in x and y , with an additional vertical translation in z . Therefore, the gantry transform may be written as

$${}^W\mathbf{T}_G = \mathbf{T}_x(x) \mathbf{T}_y(y) \mathbf{T}_z(z), \quad (16)$$

with

$$\begin{aligned} \mathbf{T}_x(x) &= \begin{bmatrix} 1 & 0 & 0 & x \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{T}_y(y) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\ \mathbf{T}_z(z) &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & z \\ 0 & 0 & 0 & 1 \end{bmatrix}. \end{aligned} \quad (17)$$

Thus, although the actuation of CoreXY is coupled at the motor level, the carriage pose remains decoupled in Cartesian task space.

c. Tool-Side Rotational Chain

The proposed architecture introduces two revolute joints on the tool side. The first joint, with angle θ_1 , is located at the connection between the tool assembly and the X-gantry. The second joint, with angle θ_2 , is located at the toolhead itself. A rigid link of length L connects the two joints. A fixed transform ${}^J_2\mathbf{T}_N$ accounts for the offset from the second joint to the nozzle tip.

A general spatial model of the tool chain can therefore be written as

$${}^G\mathbf{T}_N = {}^G\mathbf{T}_{J_1} \mathbf{R}_{\mathbf{u}_1}(\theta_1) \mathbf{T}_L \mathbf{R}_{\mathbf{u}_2}(\theta_2) {}^J_2\mathbf{T}_N, \quad (18)$$

where:

- $\mathbf{R}_{\mathbf{u}_1}(\theta_1)$ is the rotation about the first joint axis,
- $\mathbf{R}_{\mathbf{u}_2}(\theta_2)$ is the rotation about the second joint axis,
- $\mathbf{T}_L$ is the rigid translation of magnitude L between the two rotary joints.

If the link is aligned with the local z -axis of $\{J_1\}$, then

$$\mathbf{T}_L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & L \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (19)$$

For a concrete example, if the first rotation is about the local x -axis and the second is about the local y -axis, then

$$\mathbf{R}_x(\theta_1) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta_1 & -\sin\theta_1 & 0 \\ 0 & \sin\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (20)$$

$$\mathbf{R}_y(\theta_2) = \begin{bmatrix} \cos\theta_2 & 0 & \sin\theta_2 & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta_2 & 0 & \cos\theta_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (21)$$

In that case, the tool chain becomes

$${}^G\mathbf{T}_N = {}^G\mathbf{T}_{J_1} \mathbf{R}_x(\theta_1) \mathbf{T}_L \mathbf{R}_y(\theta_2) {}^J_2\mathbf{T}_N. \quad (22)$$

d. Combined Forward Kinematics

Combining the Cartesian base motion and the articulated tool chain yields the full nozzle pose in the world frame:

$${}^W\mathbf{T}_N = {}^W\mathbf{T}_G {}^G\mathbf{T}_{J_1} \mathbf{R}_{\mathbf{u}_1}(\theta_1) \mathbf{T}_L \mathbf{R}_{\mathbf{u}_2}(\theta_2) {}^J_2\mathbf{T}_N. \quad (23)$$

Substituting (23) into (15) gives the nozzle pose relative to the part:

$${}^P\mathbf{T}_N = ({}^W\mathbf{T}_P)^{-1} {}^W\mathbf{T}_G {}^G\mathbf{T}_{J_1} \mathbf{R}_{\mathbf{u}_1}(\theta_1) \mathbf{T}_L \mathbf{R}_{\mathbf{u}_2}(\theta_2) {}^J_2\mathbf{T}_N. \quad (24)$$

This expression provides the basic forward-kinematic model of the proposed architecture. It shows that the final nozzle pose depends on both the Cartesian carriage location and the distributed rotational configuration of the tool chain.

e. Kinematic Implications

The presence of a finite link length L between the two rotational joints has important kinematic consequences. First, angular motion at the proximal joint θ_1 produces both orientation change and translational displacement of the distal toolhead due to the lever-arm effect of the intermediate link. Second, the second joint θ_2 changes the local nozzle orientation relative to the already-rotated intermediate frame, creating a coupled spatial motion that is more expressive than a fixed vertical extrusion axis. Third, because the rotational DOF are located on the tool side rather than the part side, the architecture can in principle reach surface-normal alignments that are inaccessible to purely rotary-stage systems.

At the same time, the distributed tool-side design introduces new challenges. The link of length L increases the sensitivity of the nozzle position to angular error, so small misalignment in either rotary joint may produce amplified spatial error at the nozzle tip. In addition, singular or near-singular configurations may arise when the two rotational axes become locally ineffective at producing distinct orientation changes, or when the articulated tool chain approaches collisions with the printed geometry. These considerations motivate a more detailed study of the Jacobian, reachable orientation workspace, and calibration sensitivity of the proposed mechanism.

V. CONCLUSIONS

The conclusions should present a review of the key points of the article with special emphasis on the analysis and discussion of the results that were made in the previous sections and in the applications or extensions of them. You should not reproduce the summary in this section or repeat paragraphs already included in another sections of the work.

VI. ACKNOWLEDGMENT

If there are acknowledgments, references to research projects, or work-financing entities, these should be included in the ‘‘Acknowledgments’’ section afterward the conclusions of the work. Check the following: if the place the names and/or codes corresponding to research projects, institutions, financing programs, etc., involved in the work.

A. APPENDICES

In some situations it is convenient to include a section of appendices with their corresponding subsections.

a. Demonstrations

Demonstrations might be included in the main text provided that their extension and complexity do not distract the reader from the most relevant contents of the work. Otherwise, it is recommended to provide only the final results in the text including the demonstrations in an appendix at the end of the work.

b. Algorithms

In the same way as in the previous case, the algorithms are not too extensive, they might be included within the main body of the work, otherwise they should be arranged in an appendix at the end of the work.

c. Other data

The inclusion of technical details, accessory measurements, data tables or other information relevant to the work should be done in the form of an appendix when justified, otherwise they can be included directly in the main body of the text when reference is made to the same.

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